

KUBERNETES IN PRODUCTION · ISSUE 01 OF 08

Requests, limits and QoS.

Requests are for the scheduler. Limits are for protection.

01

AR

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Swipe →

What each setting does

01 Request

The scheduler reserves this much on a node.

02 CPU limit

The container is throttled above it.

03 Memory limit

The container is OOMKilled above it.

04 Nothing set

BestEffort: first to be evicted under pressure.

QoS classes

CLASS	WHEN	EVICTON
Guaranteed	requests = limits, every container	Last
Burstable	Some requests or limits set	Middle
BestEffort	Nothing set	First

Sizing habits

DO

- ✓ Set requests from observed p95
- ✓ Memory limit = request for steady apps
- ✓ Use VPA recommendations as input
- ✓ Revisit after big releases

AVOID

- ✗ 1 CPU / 1Gi everywhere
- ✗ Tight CPU limits on latency-critical paths
- ✗ Memory limit below startup peak
- ✗ No requests at all

Check before you guess

BASH

```
kubectl top pods -n payments --containers
```

```
kubectl get pod api-7d9 -n payments \
  -o jsonpath='{.status.qosClass}'
```

```
kubectl describe pod api-7d9 -n payments \
  | grep -A4 "Last State"    # OOMKilled?
```

Throttling shows in `container_cpu_cfs_throttled_periods_total`, not in `kubectl top`.



How I size a new service

- 1 Load test
- 2 Observe p95 usage
- 3 Set requests
- 4 Set memory limit
- 5 Watch throttling + OOMs
- 6 Revisit monthly

Sizing is a loop, not a one-time YAML edit.

KEY TAKEAWAY

Size from data, then keep watching.

- Requests drive scheduling and HPA math
- Memory limits kill, CPU limits throttle
- Know your QoS class

YOUR TURN

Do you set CPU limits on latency-sensitive services, or only requests?

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